

VM-22 深度调研报告

北美年金原则导向准备金落地实施全景分析

VM-22 Deep Dive: Principle-Based Reserving for Annuities
Full Implementation Landscape

Role Perspective: Chief Actuary / ALM Head
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Sources: Milliman, Oliver Wyman, PwC, EY, SOA, NAIC, Aon

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一、实施落地挑战：FRA / FIA / SPIA

Implementation Challenges: FRA / FIA / SPIA

1.1 总体框架转变

Framework Shift

VM-22 于 2026年1月1日正式生效，取代沿用数十年的 CARVM 法定准备金方法。新框架要求保险公司对非可变年金采用原则导向法（PBR）。核心变化包括：

VM-22 took effect January 1, 2026, replacing the decades-old CARVM. Key changes:

- 随机准备金（SR）：基于数千个真实世界经济情景，计算 CTE70（最坏 30% 情景的平均值）
Stochastic Reserve (SR): CTE70 across thousands of real-world economic scenarios [1][3]
- 确定性准备金（DR）：单一确定性情景下的准备金，适用于通过排除测试的产品
Deterministic Reserve (DR): Single-scenario reserve, when exclusion test is passed [3]
- 孰高原则：最终准备金 = max(SR, DR)，受 SPA 披露约束
Final Reserve = max(SR, DR), subject to SPA disclosure [1][8]

Stochastic Exclusion Test (SET) 与 Single Scenario Test (SST) 决定每条产品线的路径：

SET and SST determine the reserving path for each product line:

Test	Purpose	Pass Condition
SET	Skip SR requirement	Behavior not economically sensitive
SST	DR can replace SR	Single scenario captures all risk

1.2 FIA（固定指数年金）— 挑战最大

Fixed Indexed Annuities — Highest Impact

- 路径依赖特性（path-dependent options）——指数累积、ratchet、lookback——要求每个随机情景中进行完整现金流跟踪，状态空间和运行时间急剧膨胀
Path-dependent options (ratchets, lookbacks) require full cashflow tracking per scenario — state space explodes [3][4]
- GLWB rider 使退保行为高度动态化，绝大多数 FIA 无法通过 SET/SST
GLWB riders make surrender behavior dynamic — most FIAs fail SET/SST [4][6]
- CTE70 机制下尾部情景直接决定准备金水平，对冲有效性对结果影响巨大
CTE70 amplifies tail scenarios — hedge effectiveness directly impacts reserve outcomes [6]

1.3 FRA / MYGA（固定利率年金）

Fixed Rate Annuities — Moderate

- 纯 payout 形式的简单 MYGA 可能通过 SST，但含退保权的递延型 MYGA 往往失败
Pure payout MYGAs may pass SST; deferred MYGAs with surrender options typically fail [1][9]
- 再投资利率假设是关键变量——不同利率路径下再投资收益差异直接影响准备金
Reinvestment rate assumption is critical — different rate paths directly impact reserves [9]
- 退保率的动态建模成为不可避免的要求
Dynamic surrender modeling becomes unavoidable [1]

1.4 SPIA（即期年金）

Immediate Annuities — Lower but Not Negligible

- 无退保权的纯 SPIA 通常可通过 SST
Pure SPIA without surrender typically passes SST [1]
- 嵌入式辅助选项或二次保证会改变路径要求，触发随机测试
Embedded ancillary options or secondary guarantees can trigger SR [9]
- 死亡率假设的审慎余量（prudent margin）成为关键变量
Mortality prudent margins become a key reserve driver

二、精算建模与技术执行

Modeling Work & Technical Execution

2.1 建模软件重构痛点

Software Migration Pain Points

行业主流精算平台（MG-ALFA, Prophet, Axis）均面临大规模重构压力：

Major platforms (MG-ALFA, Prophet, Axis) face significant restructuring: [1][4]

Platform	Core Challenge
MG-ALFA	Cashflow engine must support GOES scenario format; structured asset modeling gaps
Prophet	Nested stochastic runtime explosion — single-node architecture insufficient for 1000s scenarios
Axis	Assumption governance module needs major upgrade for annual review and audit trail

2.2 嵌套随机模拟与运行时间缩减

Nested Stochastic & Run-time Reduction

VM-22 核心计算量 = N（情景数）x T（投影年数）x P（保单数）。行业主要应对策略：

Core computation = N(scenarios) x T(years) x P(policies). Industry strategies: [1][4]

- 保单聚类（Policy Cluster Modeling）：相似保单分组为 model points，降维 90-95%；聚类偏误需通过敏感性测试验证
Policy clustering reduces dimensionality 90-95%; bias must be validated via sensitivity testing [4]
- 情景缩减（Scenario Reduction）：NAIC 允许，前提是有证据表明不会实质性地降低 DR 或 SR；常用 PCA 和重要性抽样
NAIC allows scenario reduction if evidence shows no material understatement of DR/SR [4]
- 高性能计算（HPC/Cloud）：本地网格计算或云端并行处理成为必选项；大型保险公司已提高计算预算 3-5 倍
On-site grid or cloud HPC is mandatory; leading insurers increased compute budgets 3-5x [7]

2.3 动态退保假设

Dynamic Lapse Assumptions

VM-22 要求退保假设随经济情景动态调整——从 CARVM 到 PBR 最剧烈的变化之一：

VM-22 requires dynamic lapse assumptions — one of the most dramatic shifts from CARVM: [2][6]

- 退保率 = f(市场利率 - 贷记利率, 退保费用, 账户价值走势)
Lapse rate = f(market rate - credited rate, surrender charges, account value trends)
- 高利率情景中退保率激增，保险公司被迫在低点变现资产
High-rate scenarios trigger lapse spikes, forcing asset sales at market lows
- 校准难点：美国行业仅积累了有限的经济压力期经验数据
Calibration challenge: limited US experience data from stressed economic periods [6]

Oliver Wyman 2024 年分析表明，2023 年利率快速上升期间固定年金退保率显著上升，传统静态假设完全无法捕捉这一动态。

Oliver Wyman (2024): 2023's rapid rate rise caused significant fixed annuity lapses — static assumptions completely missed this dynamic. [6]

三、资产负债管理（ALM）考量

Asset Liability Management Considerations

3.1 对冲策略与 CDHS 要求

Hedging & Clearly Defined Hedging Strategy

VM-22 将对冲策略的认可度（CDHS）提升到全新监管高度：

VM-22 elevates hedging recognition (CDHS) to a new regulatory level: [8][3]

Requirement	Details
Regulatory Approval	CDHS must be approved by domiciliary state commissioner before special hedge treatment is allowed [8]
Documentation	Hedging strategy must be formally documented and consistently applied [3]
Effectiveness Testing	Hedge effectiveness testing with back-testing evidence required [3]
GLWB Hedging	FIA GLWB rider hedges need explicit effectiveness modeling and cost evidence [1]

- 如果声称对冲但未满足 CDHS，准备金按未对冲状态计算——波动市况下差异巨大
Claiming hedging without CDHS = reserves calculated as unhedged — massive difference in volatile markets [1]
- Milliman：是否承认未来对冲策略是 VM-22 实施中最早需要做出的决策之一
Milliman: Whether to recognize a future hedging strategy is among the earliest VM-22 decisions [1]

3.2 久期匹配与资产配置

Duration Matching & Asset Allocation

VM-22 明确鼓励保险公司匹配资产与负债的久期或现金流：

VM-22 explicitly encourages duration/cashflow matching of assets and liabilities: [5][9]

- 错配成本在尾部经济情景中被放大——极端情景下的错配损失直接推高 CTE70
Mismatch costs are amplified in tail scenarios — directly increasing CTE70
- 传统 CARVM 不要求明确的久期匹配；VM-22 将 ALM 从软性管理变为硬性准备金约束
CARVM had no explicit duration matching requirement; VM-22 makes ALM a hard reserve constraint [9]

3.3 私募信贷与非流动性资产

Private Credit & Illiquid Assets

- 私募信贷的浮动收益率在 2025 年具吸引力，但 VM-22 要求定制化现金流建模
Private credit's floating yield is attractive but requires bespoke cashflow modeling under VM-22 [2]
- 结构化资产（MBS, CMBS, CLO）含嵌入式期权，需比传统平台更细致的处理
Structured assets with embedded options require more granular treatment than legacy platforms provide [1]
- Oliver Wyman (Feb 2026)：VM-22 迫使年金承保人重新思考资产优化策略
Oliver Wyman (Feb 2026): VM-22 forces annuity writers to rethink asset optimization [9]

四、市场影响与产品重新定价

Market Impact & Product Repricing

4.1 资本影响量化估算

Capital Impact Estimates

公开的行业量化估算仍有限且高度依赖产品设计。以下是可获得的趋势判断：

Public estimates remain limited and highly product-design-dependent. Observed trends:

Product	Capital Direction	Key Drivers
MYGA	Neutral to mild strain	Dynamic surrender, reinvestment rate paths
FIA w/ GLWB	Potential relief vs CARVM, or higher volatility	Asset modeling, hedging, reinsurance structure [8]

RILA Under reassessment Some firms reposition as capital-efficient alternative [1]

VM-22 对资本的影响不是统一的——取决于每家公司的资产配置、对冲策略、再保险安排和产品设计。

VM-22’s capital impact is not uniform — it depends on each firm’s asset allocation, hedging, reinsurance, and product design.

4.2 产品重新定价

Product Repricing

行业正在发生的反应：

Industry-wide responses: [4][9]

- 紧缩贷记利率：MYGA 和 FIA 的封顶/参与率正在被调整
Tightening credited rates: MYGA and FIA caps/participation rates are being adjusted
- 缩短保证期：降低长期保证带来的尾部风险
Shortening guarantee periods: reducing tail risk from long-term guarantees
- 重设计 rider：GLWB 收费上调或特征简化
Redesigning riders: GLWB fee increases or feature simplification
- 调整退保费用表：减少早期退保对资本的压力
Adjusting surrender schedules: reducing capital strain from early lapses

4.3 离岸再保险浪潮

Offshore Reinsurance Wave

- 美国寿险行业离岸再保险规模已超过 1 万亿美元，百慕大、开曼群岛为主要接收方
US life offshore reinsurance exceeds \$1 trillion, primarily to Bermuda and Cayman [2][6]
- VM-22 要求再保险前后的准备金单独计算，并添加交易对手违约边际——降低离岸再保险的资本释放效果
VM-22 requires pre/post-reinsurance reserve calculation + counterparty default margin — reducing offshore capital relief [8]
- 2025-2026 年 NAIC 和监管机构正加强对再保险风险转移真实性的审查
NAIC/regulators intensifying scrutiny of reinsurance risk-transfer真实性 in 2025-2026 [3]

五、精算工作量与运营冲击

Actuarial Workload & Operational Impact

5.1 人力与系统瓶颈

Staffing & System Bottlenecks

根据 PwC 和 SOA 的行业反馈：

Industry feedback from PwC and SOA: [1][4][7]

Area	Impact Level	Details
Actuarial Team	Significant expansion	Model dev, assumption setting, VM-31 reporting: 20-50% headcount increase
IT / Data Team	Major increase	Data pipelines, HPC, GOES integration need system architects
Risk Management	Moderate increase	ALM, hedge effectiveness, stress testing need more analysts
External Consultants	Short-term surge	2024-2026: record VM-22 consulting revenue at Milliman, PwC, EY

5.2 VM-31 报告负担

VM-31 Reporting Burden

VM-31 PBR Actuarial Report 的要求远超 CARVM 报告：

VM-31 requirements far exceed CARVM reporting: [3][4]

- 方法论描述、假设设定及依据、对冲策略说明
Methodology descriptions, assumption rationale, hedging strategy documentation
- 复杂资产建模方式、情景测试和缩减方法
Complex asset modeling methods, scenario testing and reduction approaches
- 聚合分组的决策依据
Aggregation subgrouping rationale
- 监管审查日趋严格——2025年10月美国精算学会 VM-31 专项网络研讨会明确审查重点
Regulatory scrutiny intensifying — Oct 2025 Academy VM-31 webinar detailed review priorities [3]

5.3 季度/年度结账冲击

Quarterly & Annual Closing Impact

Period	Milestone
2024-2025	Preparation: gap assessment, model rebuild, assumption governance framework
2026 Q1+	VM-22 effective for new business; full stochastic reserve run each quarter
Year-end 2026	First SPA disclosure (not yet reserve floor) [8]
2029	Mandatory compliance: all in-force blocks under PBR [5]

实操痛点：从基于 Excel/历史公式的 CARVM 到重度建模的 PBR，季度结账时间预计延长 2-4 周，直到流程和自动化成熟。

Pain point: CARVM-to-PBR transition expected to extend quarterly close by 2-4 weeks until processes and automation mature. [7]

关键结论

Key Conclusions

1. VM-22 不是准备金方法变更，而是运营模式变革

VM-22 is not a reserving change — it is an operating model transformation requiring cross-functional coordination across actuarial, IT, ALM, risk, and finance.

2. FIA 是变革风暴中心

FIAs are at the epicenter: path dependency + GLWB + hedging complexity create the greatest challenge.

3. 建模基础设施是最严重瓶颈

Modeling infrastructure is the #1 bottleneck: nested stochastic compute demands, GOES integration, data pipeline construction far exceed industry expectations.

4. ALM 从软约束变为硬约束

ALM evolves from soft management tool to hard reserve constraint — CDHS hedge recognition and CTE70 tail-risk amplification reshape ALM decisions.

5. 精算团队规模和质量要求双升

Actuarial teams need 20-50% more headcount and upgraded skills in modeling, governance, and VM-31 reporting.

6. 再保险经济性正在被重估

Offshore reinsurance capital relief may diminish under VM-22's pre/post-reinsurance calculation and counterparty default margin requirements.

7. 2026-2029 是过渡窗口期

Companies that complete modeling and governance upgrades early will gain competitive advantage.

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